

SOFTWARE SUBSYSTEM

Nios V processor

AXI4 instruction/data
managers

JTAG UART

Timer /
cycle counter

Platform Designer interconnect

AXI/Avalon adaptation,
arbitration, address decoding

External shared SDRAM

left/right frames, best-cost
state, disparity output

CPU memory traffic

shared memory path

DOMAIN ACCELERATOR

Control/status slave

base addresses, dimensions,
start/done, counters

Avalon-MM memory master

burst reader and writer

Stereo SAD pipeline

line buffers, P disparity
lanes, winner-take-all

benchmark control

status / counters

register access

burst reads/writes

data/bus path

control path

Shared-memory rule: both masters access the same SDRAM address space. Internal FIFOs and line buffers retain only transient in-flight data; they are not CPU-managed private frame stores.